How Foveated Vision Shapes Cognitive Maps in Navigation Agents

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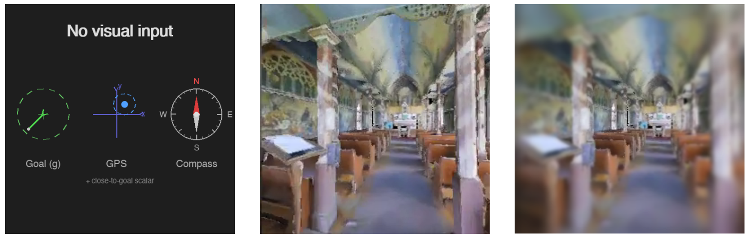
Abstract

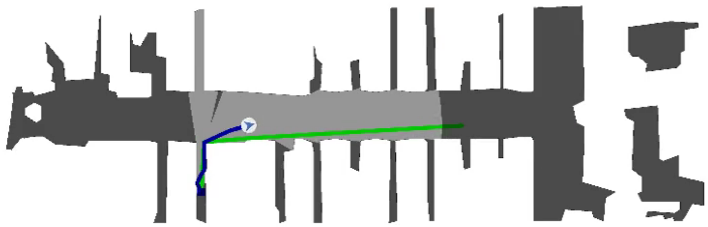
What shapes the content of a recurrent navigation agent's learned internal state — the task, the architecture, or the structure of the visual input? Prior work has shown that blind PointNav agents spontaneously develop cognitive-map-like representations in their recurrent state under task pressure alone [Wijmans et al., 2023]. We show that the blind result is a special case of a broader principle: *the visual encoder's information capacity determines whether the LSTM has to carry world-frame spatial structure*. Holding task, optimiser, and LSTM backbone identical and varying only the visual pathway (no vision; a resolution-collapsed encoder; uniform vision; foveated vision at a fixed gaze; foveated vision with a learned gaze), we probe the recurrent state for world-frame position and heading. When the encoder cannot resolve those quantities, the LSTM develops a persistent, integration-capable code for them; when it can, the recurrent state drifts to pass-through. We call this the *encoder–memory race*; it recasts the "cognitive maps emerge without vision" story as "cognitive maps emerge when the visual pathway cannot resolve the map on its own." A second finding cuts across conditions: five agents that solve the task equally well develop recurrent representations in essentially orthogonal linear subspaces, and cross-condition memory transplants incur task cost well above transplant mechanics — sensor structure shapes representational *format*, not only strength. A third question — whether static gaze location within a rich-encoder regime acts as a second content axis — is tested by a foveated-shifted causal control whose training is in progress. The H1 ordering reproduces on a held-out MP3D evaluation, consistent with an agent-internal rather than Gibson-specific mechanism. Three implications, each scoped to the recurrent PointNav setting we study. **(i)** Encoder capacity correlates inversely with linear, top-layer LSTM spatial encoding; *if* the correlation reflects causation (which the encoder-resolution scaling sweep, in progress, tests directly), it would also be a practical training-time lever for inducing cognitive-map-like internal representations. **(ii)** "More pixels, richer memory" is the wrong intuition within this setup; the relationship across our five conditions is inverse. **(iii)** The pattern parallels independent animal-navigation findings — enhanced hippocampal coupling in congenitally blind humans, the disruption of rodent grid-cell firing in the dark (with navigation nonetheless preserved), cross-modality hippocampal remapping in bats — without depending on their biophysics, suggestive of a sensor–memory coupling principle whose generality across architectures (transformers, non-RL learners, non-navigation tasks) we do not test here.

1 Introduction

Recurrent navigation agents trained with deep RL build internal states that support memory, planning, and generalisation, and the content of those internal states is at least as interesting as the task metrics they drive. A natural question is what *determines* that content: architecture, objective, training budget,



(a) Blind (b) Uniform (c) Foveated (fixed gaze)



(d) Bird's-eye navigation map: trajectory (blue) vs. geodesic shortest path (green).

Figure 1: The five visual conditions differ only in what reaches the agent's encoder. (a) Blind: no visual input; only the non-visual sensor stack (goal-in-start-frame, GPS, compass, close-to-goal scalar) — the Wijmans [Wijmans et al., 2023] configuration. (b) Uniform: 256×256 full-resolution RGB. (c) Foveated (fixed gaze): eccentricity-dependent Gaussian blur centred on the image middle ($\sigma_{\max} = 8$, quadratic falloff). The two additional conditions not shown are *matched-compute* (a down-sampled uniform view with the same total pixel budget as foveated) and *foveated (learned gaze)*, which adds a learned (x, y) gaze offset to (c). (d) All conditions solve PointGoal navigation in the same Habitat Matterport3D / Gibson scenes; the top-down view is shown for reference only and is not available to the agent.

or visual input? Wijmans et al. [Wijmans et al., 2023] showed that map-like spatial representations emerge from task pressure alone, even in blind agents that receive only goal displacement and proprioception; Dwivedi et al. [Dwivedi et al., 2022] probed sighted agents for related concepts; Banino et al. [Banino et al., 2018] and Cueva & Wei [Cueva and Wei, 2018] demonstrated grid-cell-like structure in path-integration networks. Together these results establish that *some* spatial structure is learned without being designed in. They leave open the complementary question that we take up here: within agents that all achieve high task success, how much of *what* recurrent memory encodes is determined by the structure of the visual input rather than by the task? In biology the answer is clear — primate hippocampal cells code joint place–view–action state because primates sample the scene with a foveated sensor [Wirth et al., 2017, Rolls, 2024], and the same bat produces non-overlapping hippocampal populations in vision versus echolocation [Geva-Sagiv et al., 2016]. We investigate the analogous question in a controlled agent setting where the only free variables are the visual input structure and the gaze that directs it.

Our five-condition design isolates three degrees of freedom: sensor type (blind / uniform / foveated), encoder capacity (uniform's full ResNet-18 vs. matched-compute's $1{\times}1$-collapsed feature map), and gaze location (fixed vs. learned). Every condition uses the same DD-PPO learner, the same Gibson-0+MP3D training pool, the same LSTM backbone, and the same non-visual sensor stack (Figure 1). We adopt the LSTM architecture from Wijmans et al. [Wijmans et al., 2023] deliberately: the emergent-map literature we build on is recurrent, and matching that architecture lets our findings be interpretable as statements about *the structure of the task-shaped memory* rather than about the recurrent-versus-transformer design choice. We discuss in §5 whether the logic plausibly extends to transformer-backed or non-navigation systems — an empirical question we do not settle here.

We test three hypotheses. **H1 (encoder–memory race).** Agents whose visual encoder cannot resolve world-frame position or heading should develop these quantities in their recurrent state; agents with a rich visual encoder should not. Signature: current-state GPS / compass probe R^2 tracks visual-encoder information content monotonically across the five conditions, with persistent decoding at lags of ≥ 10 time-steps. **H2 (sensor structure → format divergence).** Different sensor conditions should produce internal representations in distinct linear subspaces, not scaled versions of a common code. Signature: pairwise CKA near zero, probe weights fail to transfer across conditions while within-condition probes succeed, and cross-condition memory transplants degrade task performance beyond same-condition self-transplants. **H3 (gaze location → orthogonal content lever).** Where a foveated sensor is centred should act as a representational-content axis independent of the sensor transform. We test this with a foveated-shifted control in which gaze is hardcoded to a different image location from the default central gaze; same foveation transform, same training budget, different gaze target. Our minimal learned-gaze module collapsed to a static output (Appendix B), so we use that collapsed location for the shifted-gaze control rather than as evidence about active gaze; the foveated-shifted training is in progress (§4.4). Each hypothesis has a direct analog in the animal-navigation literature [Kupers and Ptito, 2014, Chen et al., 2016, Geva-Sagiv et al., 2016, Jutras et al., 2013, Najemnik and Geisler, 2005], which we return to in Discussion as evidence that the effects reflect general sensor–memory coupling principles rather than architectural artefacts.

Findings H1 is supported by a clean monotonic effect: LSTM world-frame GPS encoding tracks visual-encoder information capacity from $R^2 = 0.95 \pm 0.02$ (blind, no encoder) and $R^2 = 0.78 \pm 0.10$ (matched-compute, 1×1 encoder) down to chance ($R^2 \in [-2.4, +0.06]$ with CV $\sigma \in [0.7, 4.0]$) for the three full-resolution-encoder conditions, with the bottleneck-conditions' code persistent at lags ≥ 20 time-steps. The same ordering reproduces on a held-out MP3D evaluation, confirming an agent-internal mechanism. H2 is supported by four independent measures of cross-condition divergence: pairwise CKA at the noise floor ($< 10^{-4}$), catastrophic probe-transfer failure ($R^2 \ll -800$ in every off-diagonal cell), perfect 1-NN cluster purity (1.000 across 7500 pooled hidden states, vs. chance 0.20), and behaviourally costly cross-condition memory transplants (SPL drops 0.19–0.21 between rich-encoder pairs, well above the self-transplant noise floor). H3 asks whether gaze location is a second content axis *within* the rich-encoder regime; the clean causal test (foveated-shifted control) is in training at submission and will populate §4.4. A non-linear (MLP) probe recovers position from the rich-encoder *top-layer* hidden states ($R^2 \in [0.51, 0.73]$), sharpening rather than weakening H1. We interpret this together with the per-layer probes (Figure 6 right): every condition has linearly-decodable GPS at LSTM Layer 0 (where the GPS sensor embedding enters the input vector — so this is at least partly a sensor-pass-through effect, not a network "computation"); the divergence is what happens through Layers 1–2. Bottleneck conditions retain linearly-accessible GPS to the top layer used by DD-PPO's linear action head; rich-encoder conditions do not. The MLP recovery at the top layer is consistent with position information persisting in a non-linear subspace, though we cannot from this alone rule out MLP exploitation of position-correlated features (wall distance, scene identifiers) rather than position itself — a question the encoder-resolution scaling sweep (Appendix D, in progress) will help resolve. H1 is most precisely stated as a claim about *linear, top-layer, policy-accessible* GPS encoding.

Contributions (i) A controlled five-condition ablation that decouples "no vision" from "visual-encoder bottleneck" via the matched-compute condition, making the encoder–memory race claim falsifiable. (ii) Convergent evidence (linear probes, MLP probes, CKA, probe transfer, 1-NN purity, memory transplant, shortcut discovery, MP3D generalisation, per-layer probes) for two of the three hypotheses, with a clean methodology disclosure including the deterministic-rollout protocol that fixes a probe-collection confound documented in §3.3. (iii) A mechanistic re-reading of Wijmans et al.'s blind-agent finding: cognitive maps emerge not because vision is absent but because the visual encoder cannot resolve the map, predicting an encoder-capacity scaling curve we begin to test in Appendix D.

2 Related Work

Cognitive maps in RL navigation agents Wijmans et al. [Wijmans et al., 2023] demonstrated that map-like representations emerge in the recurrent memories of DD-PPO agents even without visual input; Dwivedi et al. [Dwivedi et al., 2022] probed sighted agents for related spatial concepts; Banino et al. [Banino et al., 2018] and Cueva & Wei [Cueva and Wei, 2018] found grid-cell-like units in recurrent networks trained on path integration; Gornet & Thomson [Gornet and Thomson, 2024] extended this via visual predictive coding. Hennig et al. [Hennig et al., 2023] showed that RL under partial observability can induce belief-state-like representations without explicit probabilistic objectives. Ramakrishnan et al. [Ramakrishnan et al., 2025] examined spatial cognition in frontier models. All these studies use either absent or spatially uniform perception; none vary input quality spatially, and none test whether different sensors produce different representational formats.

Foveated vision and gaze policies in deep learning Foveated rendering [Strasburger et al., 2011, Rosenholtz, 2016] exploits eccentricity-dependent acuity falloff for efficiency. Learned glimpse policies appear in recurrent attention models [Mnih et al., 2014]; Atanov et al. [Atanov et al., 2024] showed that sensor layout changes downstream task performance. Pourrahimi & Bashivan [Pourrahimi and Bashivan, 2025] probed a foveated visual-search model for neuroscience-predicted features; Shang & Ryoo [Shang and Ryoo, 2023] jointly learned gaze and motor policies following the Sprague & Ballard [Sprague and Ballard, 2003] framework. These address visual search or task performance; none asks how sensor structure reshapes the learned spatial memory of a downstream navigation policy.

Probing neural representations Belinkov [Belinkov, 2022] surveys probing classifiers; Hewitt & Liang [Hewitt and Liang, 2019] introduced control tasks to distinguish genuine structure from probe memorisation; Kornblith et al. [Kornblith et al., 2019] introduced linear CKA for cross-model similarity. We adopt linear probing, the Hewitt–Liang selectivity metric, and unaligned linear CKA, and complement them with two behavioural interventions directly on the recurrent state (memory transplant and shortcut discovery) to validate the probe-based findings outside the linear-probe framework.

Biological precedent Our hypotheses and framing draw on three threads of animal-navigation research. (i) Primate hippocampal cells code joint place–view–action state [Wirth et al., 2017, Rolls, 2023]; Rolls [Rolls, 2024] argues that primate *spatial-view cells* and rodent *place cells* reflect the different sampling statistics of foveated versus near-panoramic vision. (ii) The same bat in the same environment produces non-overlapping hippocampal populations when it switches between vision and echolocation [Geva-Sagiv et al., 2016]—format-level divergence across sensor modalities within a single animal. (iii) In primates, each saccade resets hippocampal 3–8 Hz rhythms and the reliability of the reset predicts subsequent memory [Jutras et al., 2013]; only overt (not covert) attention automatically writes to visual working memory [Tas et al., 2016]. Gaze targets themselves are selected in an information-gain-maximising fashion consistent with ideal-observer predictions [Najemnik and Geisler, 2005, Gottlieb et al., 2013, Hayhoe and Matthis, 2018]. We use these results only to frame our ML predictions and interpret our findings; our contribution is the controlled ablation in a learned agent that cross-species and cross-modality biological comparisons cannot cleanly supply.

What is new here Three things distinguish this work from the studies above. (i) A *controlled* five-condition ablation with everything but the visual pathway held identical, including the matched-compute condition that decouples "no vision" from "visual encoder bottleneck" — the lever that makes the encoder–memory race claim falsifiable. (ii) A protocol-corrected probing methodology: deterministic action selection at probe time (the policy used by every other component of our eval pipeline) rather than stochastic sampling, which we found inflates probe R^2 for high-entropy policies via a 4-step quasi-static-trajectory artefact (§3.3, Sampling protocol). (iii) Convergent behavioural validation of the probe-based H1 and H2 findings via memory transplant and shortcut discovery, both of which sit *outside* the linear-probe framework.

3 Methods

3.1 Agent Architecture and Training

All agents are trained on PointGoal navigation in Habitat [Savva et al., 2019] using DD-PPO [Wijmans et al., 2020] on a merged pool of Gibson-0+ [Xia et al., 2018] (411 scenes) and Matterport3D train (61 scenes), totalling 472 training scenes. Evaluation uses 18 held-out MP3D test scenes (1008 episodes).

All agents share an identical 3-layer LSTM (512-d) backbone and the Wijmans non-visual sensor stack: goal-in-start-frame, GPS, compass, and a close-to-goal scalar, each projected to 32-d and concatenated with a 32-d previous-action embedding. All sighted conditions are trained for a target of 250 M environment frames; the blind baseline is trained to 342 M frames to allow slower convergence from proprioception alone.

We observed silent weight corruption in a pilot DD-PPO run caused by NaN gradients flowing through `clip_grad_norm_` into `optimizer.step()`; we mitigate this with a gradient-sanitisation patch to `PPO.before_step`, described in Appendix C.

3.2 Visual Conditions

Five conditions vary only the visual input (Figure 1):

1. *Blind:* no visual input; non-visual sensor stack only, replicating Wijmans et al. [Wijmans et al., 2023].
2. *Uniform:* full-resolution 256×256 RGB encoded by ResNet-18.
3. *Foveated (fixed):* eccentricity-dependent Gaussian blur ($\sigma_{\max} = 8$, quadratic falloff) with gaze locked at image centre; same ResNet-18 encoder.
4. *Matched-compute:* 48×48 uniform RGB. At this resolution, ResNet-18's 32-fold spatial downsampling collapses the feature map to 1×1, removing the spatial dimensions of the encoder's output (channel information is preserved as a flat 512-d vector). **[TODO: Verify whether channel info alone in matched-compute encodes any usable position signal — a small probe on the encoder feature map (not LSTM state) would settle this.]**
5. *Foveated (learned):* same blur model, but gaze centre predicted by a lightweight MLP trained end-to-end with the navigation policy.

The matched-compute condition is central to H1: its 1×1 encoder feature map provides a visual observation while leaving the encoder unable to resolve world-frame position from the current frame, separating the "no vision" account from the "encoder bottleneck" account. We attribute its strong LSTM GPS encoding to the feature-map collapse specifically, but the 48×48 input also has lower spatial resolution than uniform's 256×256, so we cannot from this single condition fully separate "1×1 feature-map collapse" from "low input resolution." The encoder-resolution scaling sweep (Appendix D, in progress) varies input resolution at fixed encoder stack to test this attribution directly.

Implementation note The foveated conditions (fixed, shifted, learned) disable the running-mean-and-variance input normaliser used by uniform and matched-compute, to avoid an in-place autograd conflict along the gaze-decoder gradient path. All reported results were obtained with this setting; we have not directly verified that re-enabling the normaliser would leave the H1 ordering unchanged. **[TODO: Verify normaliser invariance: rerun foveated-fixed with the normaliser re-enabled (the autograd conflict is gaze-decoder-specific, so feasible for fov-fix), and confirm the H1 GPS R^2 does not move.]**

3.3 Probing Methodology

After training, we freeze all weights and collect per-step LSTM hidden states (top-layer $\mathbf{h}$, 512-d) paired with ground-truth spatial labels from $n{=}500$ Gibson held-out validation episodes (the split not used in training). Rollouts use *deterministic* (argmax) action selection, matching the eval protocol of our shortcut-discovery, memory-transplant, and standard habitat-baselines evaluators. Under this protocol the rollout lengths span roughly $\sim$100–400 mean steps depending on condition (longer for blind, shorter for high-SPL conditions), so we use 5-fold episode-level CV to keep the train/test splits balanced per scene rather than length-matching across conditions. **[TODO: A controlled length-matching ablation (truncate every condition to a common length, re-run probes) is left for follow-up; we expect it to leave the H1 ordering unchanged because the bottleneck conditions decode at high R^2 even at the smallest sub-sample, but have not verified it.]** We train Ridge regression probes ($\alpha = 10$) with episode-level 5-fold cross-validation, reporting mean $\pm\sigma$ across folds, and verify probe specificity with Hewitt–Liang label-permutation selectivity [Hewitt and Liang, 2019].

Probed variables: GPS position (episodic), compass heading (episodic), distance-to-goal, lag-k path history (target value at $t - k$ decoded from $\mathbf{h}_t$, $k \in \{0, \ldots, 20\}$), visited-region occupancy, full occupancy grid (20×20), per-unit spatial information (place-cell analysis), and the ego-frame goal vector $\mathbf{R}(-\theta)(\mathbf{g} - \mathbf{p})$ decomposed into scalar goal-distance and 2-D direction.

Sampling protocol An earlier draft of this work used stochastic (policy-distribution) action selection for probe rollouts. Under stochastic sampling, conditions with higher action-entropy (specifically the rich-encoder conditions foveated, uniform, matched, blind under their trained stochastic policies) sampled the STOP action early with probability ~ 0.25 per step, producing mean episode lengths of ~ 4 steps and target variance two orders of magnitude below the episodic range. The resulting probe R^2 values were trivially high across all conditions (any predictor fits a near-constant target). This confounded cross-condition comparisons, and we switched to the deterministic protocol used here and elsewhere in the probe/eval pipeline. All results in this paper use the deterministic protocol.

For foveated-fixed, all reported analyses (probes, behavioural, eval) use checkpoint ckpt.36 at ∼174M frames, the last clean intermediate before the silent-corruption window (Appendix C).

3.4 Cross-Condition Analysis

To test H2, we employ two complementary methods. *Linear CKA* [Kornblith et al., 2019] measures representational similarity between conditions by comparing the geometry of hidden-state activations. Values near 1 indicate identical geometry; near 0 indicates unrelated representations. *Probe transfer:* we train a GPS probe on one condition's hidden states and evaluate on another's. If spatial information is encoded in a shared linear subspace, cross-condition transfer should succeed. Catastrophic failure indicates fundamentally different encoding formats. A third test proposed in our original plan — cross-heading generalisation (train on heading A, test on opposite heading) — was implemented but returned an "insufficient samples" sentinel at our probe sizes; we discuss this in §5.5.

3.5 Behavioural Probes

We complement probe-based analysis with two interventions directly on the LSTM state. *Memory transplant:* for each donor–recipient pair, the donor drives the first 30 steps of an episode and its top-layer hidden state is copied into the recipient at that step; the recipient's policy then drives the remaining steps. We compare *baseline* (recipient alone, full episode), *self-transplant* (recipient as its own donor, controls for transplant mechanics), and *cross-transplant* (different donor). Midpoint 30 was chosen because Gibson PointNav episodes average ∼100–120 steps; a later midpoint left too few steps after transplant. We use 150 episodes per pair per condition, sampled uniformly from the held-out Gibson validation split. *Shortcut discovery:* for each condition, we run episodes in pairs of two using the same scene but different goals, sampling 20 scenes with 10 paired episodes per scene (200 paired runs per condition per setting). Per-condition SPL is reported as the mean of the 20 scene-level mean SPLs (prevents high-episode-count scenes from dominating). We compare *reset* (LSTM zeroed between the two episodes) vs. *persistent* (LSTM state carried over) on the second-episode SPL. A reduction under the persistent setting measures how tightly the stored representation is locked to the previous goal. For the foveated-fixed condition, all probes and behavioural runs use the ckpt.36 checkpoint (∼174M frames, the last clean intermediate before the silent-corruption window, §4).

4 Results

4.1 Summary of Conditions and Key Metrics

Table 1 collects the per-condition numbers this section returns to. All sighted conditions solve PointGoal at 96–99% success; blind reaches 93% at a longer budget, reproducing Wijmans et al. [Wijmans et al., 2023]. The narrow success-rate spread makes "the agents differ in what they do" an unlikely explanation for the encoded-content gap we report (though it does not strictly rule it out). The probe machinery itself is calibrated: distance-to-goal decodes robustly across all five conditions ($R^2 \in [0.81, 0.90]$), confirming that Ridge has the capacity to fit high-R^2 scalars when the hidden state contains them; and Hewitt–Liang label-permutation selectivity is high in the conditions where the GPS code itself is high (blind GPS / compass selectivity $+0.95/+0.81$; matched $+0.72/+0.56$), consistent with a genuine signal rather than label memorisation. The headline pattern previewed in the table is monotonic in visual-encoder information: conditions with no encoder (blind) or a 1×1-collapsed encoder (matched) robustly encode world-frame GPS and compass in LSTM hidden state (GPS $R^2 = 0.95, 0.78$; compass $R^2 = 0.81, 0.64$); conditions with a full-resolution encoder (uniform, foveated, foveated-learned) do not (GPS $R^2 \in [-2.4, +0.06]$ with CV $\sigma \in [0.7, 4.0]$, i.e. chance). We develop this as H1. Two training-stability caveats are disclosed in full in Appendix C: foveated-fixed uses ckpt.36 at ∼174M frames (last clean checkpoint before a silent NaN-gradient corruption at ≈ 182M), and matched-compute's 1×1 feature-map collapse, originally noticed as

Table 1: Per-condition summary. Ridge $\alpha = 10$ probes on hidden states collected under deterministic argmax-action rollouts (n=500 held-out Gibson episodes per condition). Probe R^2 columns report mean $\pm$ std across 5-fold episode-level CV. *GPS* and *Compass* are world-frame; *DtG* is ego-relative distance-to-goal. Lag-5 GPS R^2 reports stability of the GPS code across 5 time-steps (persistent-integration signature). Bold entries mark the top within the H1-relevant ordering (blind $>$ matched $\gg$ others).

	Training			Current-state probes (R^2)			H1	H3	Shortcut	MP3D
Condition	Frames	SPL	Succ.	GPS	Compass	DtG	lag-5 GPS	goal-dir R^2	rel. drop %	ΔGPS
Blind	342M	0.59	0.93	**+0.95±0.02**	**+0.81±0.08**	+0.90±0.03	**+0.95±0.02**	−0.06	52	−0.02
Matched-compute	250M	0.67	0.98	**+0.78±0.10**	**+0.64±0.10**	+0.85±0.12	**+0.78±0.10**	−0.03	18	−0.07
Uniform	250M	**0.85**	**0.99**	−0.31±0.86	+0.36±0.23	+0.86±0.09	−0.27±0.85	−0.03	41	−0.04
Foveated (fix)	174M	0.78	0.96	+0.06±0.88	+0.07±0.69	+0.82±0.09	+0.04±0.93	−0.08	21	+0.30
Fov-learned	250M	0.82	0.98	−2.43±3.98	−1.34±3.14	+0.81±0.09	−3.60±6.20	**[TODO: H3]**	15	+1.65

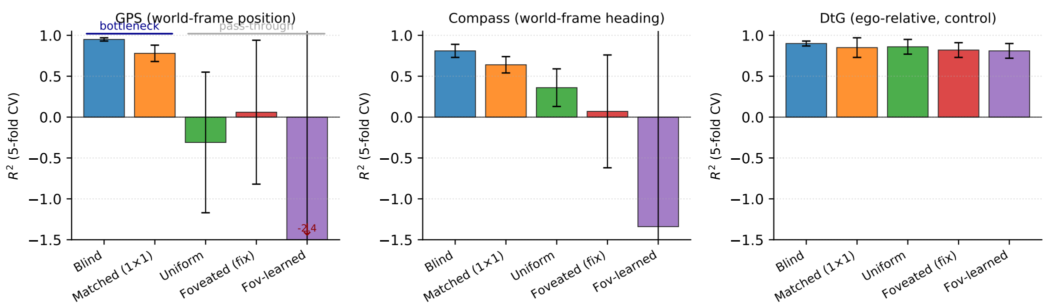


Figure 2: Current-state probe R^2 (5-fold CV, episode-level splits, Ridge $\alpha = 10$ on deterministic-rollout hidden states) across five conditions. *GPS* (world-frame position) and *compass* (world-frame heading) show a clean monotonic ordering tracking visual-encoder information content: blind (no visual input) and matched-compute (48×48 RGB $\to$ 1×1 feature map) develop rich world-frame representations in their recurrent state; uniform, foveated, and foveated-learned (full 256×256 RGB) do not. *DtG* (distance-to-goal, an ego-relative task quantity) decodes robustly in *all* conditions because every agent must track it to navigate. Error bars: $\pm 1\sigma$ across 5 episode-level CV folds.

a training quirk, is what makes matched-compute a useful test of the encoder-bottleneck framing (though we cannot from one condition alone fully separate the 1×1 feature-map effect from the lower input resolution; see §3.2 and Appendix D).

4.2 H1: Visual Encoder Bottleneck → LSTM Spatial Compensation

Finding Visual-encoder capacity inversely tracks LSTM spatial-encoding content across our five conditions. Blind agents encode current GPS in the recurrent state at $R^2 = 0.95 \pm 0.02$; matched-compute agents (48×48 RGB input, 1×1 encoder feature map) at $R^2 = 0.78 \pm 0.10$; all three full-resolution-encoder conditions (uniform, foveated, foveated-learned) decode GPS at chance ($R^2 \in [-2.4, +0.06]$ with CV $\sigma \in [0.7, 4.0]$). Compass shows the same monotonic ordering at a smaller effect size ($0.81 \to 0.64 \to 0.36 \to 0.07 \to -1.3$). DtG decodes robustly everywhere ($R^2 \in [0.81, 0.90]$). Figure 2 visualises the pattern. **[TODO: Whether this correlation reflects causation (encoder capacity *drives* LSTM encoding) is what the encoder-resolution scaling sweep, Appendix D, tests directly. Multi-seed runs of the existing five conditions are also in progress to bound seed-to-seed variability.]**

Integration signature at lag k The bottleneck conditions' GPS encoding is *persistent*, not an instantaneous readout. Decoding GPS_{t-k} from $\mathbf{h}_t$ yields $R^2 = 0.95$ at k=0, 2, 5, 10 for blind and $R^2 = 0.78$ at every k up to 10 for matched (Table 1). Pushing k further on a separate run, blind GPS still decodes at $R^2 = 0.92$ at $k = 20$; matched holds above $R^2 = 0.50$ at $k = 20$ (with smaller n_{test} as short episodes drop out at high lag). The rich-encoder conditions decode at chance at every $k \leq 10$. Compass decays steadily with k in both bottleneck conditions ($0.81 \to 0.19$ over 10 steps

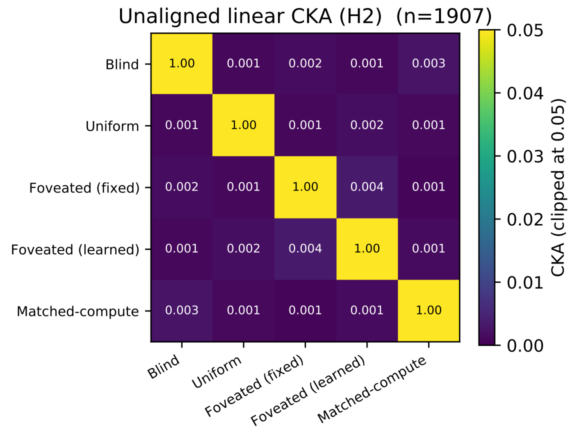


Figure 3: Unaligned linear CKA [Kornblith et al., 2019] between top-layer LSTM hidden states (n=95,640 steps per condition under deterministic-rollout probes). Every off-diagonal is below 10^{-4}; cross-condition similarity is at the numerical-noise floor. Within-condition split-half CKA (not shown) is ≈ 1, confirming the near-zero off-diagonals reflect representational divergence, not estimator noise.

for blind; $0.64 \to -0.16$ for matched) but never becomes available in rich-encoder conditions. This is the integration signature of a compensatory cognitive-map code: the LSTM maintains a stable world-frame position representation across many time steps, precisely because the visual encoder cannot do so on its own.

Proposed mechanism We *propose* the following account, consistent with the data but not directly tested. In PointNav, GPS and compass are observations that the policy reads each step; the recurrent state need only re-encode them if it cannot rely on the visual stream to do so. A full-resolution RGB encoder can re-derive position and heading at every step. An impoverished encoder cannot, and we suggest that the policy-optimisation pressure to produce trajectory-consistent actions selects for an LSTM that carries the missing variables. Blind and matched-compute bracket this picture: blind has *no* visual encoder (pure compensation); matched has a *collapsed* 1×1 encoder (partial compensation); foveated, uniform, and foveated-learned all have full encoders (pass-through). Task success is tightly bunched (0.59–0.85 SPL), so a skill-gap artefact is unlikely to explain the encoding gap. **[TODO: A direct causal test of the proposed mechanism — e.g., training a rich-encoder agent with GPS perturbed mid-rollout, or training a bottleneck agent with the GPS sensor removed — is left for future work.]**

Relation to Wijmans et al. Wijmans et al. [Wijmans et al., 2023] showed that blind PointNav agents develop spatial representations in LSTM memory. Our blind result ($R^2 = 0.95$) replicates this; matched-compute extends it beyond the blind special case to a condition that has visual input but cannot resolve world-frame structure from it. We interpret this as evidence that the relevant trigger is the visual encoder's information capacity, not the binary presence/absence of vision. Appendix D reports an encoder-resolution sweep (32×32 to 256×256; in progress) which tests the trigger directly as a scaling curve.

Bio parallel The pattern parallels the animal-navigation literature without depending on its mechanisms: congenitally blind humans develop enhanced hippocampal coupling and outperform sighted controls on non-visual spatial tasks [Kupers and Ptito, 2014]; rodent grid cells lose their hexagonal periodicity in darkness even with head-direction input intact [Chen et al., 2016], yet the animal still navigates — so spatial bookkeeping that the (vision-dependent) grid code can no longer carry has to come from somewhere else. We use these only as analogy: in both cases visual-pathway absence shifts spatial bookkeeping into a non-visual substrate, the same qualitative direction we measure in LSTM hidden states. We do not claim the agent and biological mechanisms are the same.

Table 2: Probe-transfer GPS R^2 on deterministic-rollout hidden states: train on row condition, test on column. Self-accuracy diagonal is 0.89–0.99; every cross-condition transfer is catastrophically negative ($R^2 \ll -800$, often $\ll -10^4$), reflecting that a probe trained on one condition's hidden-state geometry does not just degrade on another — it predicts off-manifold values.

Train \ Test	Blind	Uniform	Fov-fix	Fov-lrn	Matched-compute
Blind	$+$**0.99**	-844	$-11{,}301$	$-10{,}731$	$-55{,}706$
Uniform	$-1{,}383$	$+$**0.90**	$-7{,}046$	$-3{,}305$	$-110{,}317$
Fov-fix	$-2{,}091$	$-2{,}792$	$+$**0.92**	$-1{,}263$	$-39{,}100$
Fov-lrn	$-6{,}722$	$-2{,}423$	$-6{,}205$	$+$**0.89**	$-38{,}842$
Matched-compute	$-10{,}621$	$-4{,}832$	$-4{,}984$	$-58{,}169$	$+$**0.96**

4.3 H2: Sensor Structure $\rightarrow$ Format-Level Divergence

Finding Five agents that solve the same task at the same success rate do not share a spatial representation. Pairwise linear similarity (CKA) sits at the noise floor for every condition pair (Figure 3; off-diagonals all $< 10^{-4}$); a GPS probe trained on one condition's hidden states and evaluated on another's does not just degrade — it predicts wildly off-manifold values, with cross-condition transfer $R^2 \ll -800$ in every cell (Table 2). In plain terms: what each agent calls "position" is written in a different linear subspace. Visual input type shapes *format*, not just strength.

Ruling out alternatives Self-probes succeed for every condition, so probe capacity is not the bottleneck. CKA is computed on a common-sample truncation, ruling out sample-size asymmetry. Task competence is tightly bunched (96–99$\%$ success across sighted conditions), so the divergence is not a skill difference. And transfer fails symmetrically (both directions), which rules out the possibility that one condition's code simply embeds another's.

A t-SNE projection of pooled hidden states confirms the CKA result visually (Appendix A): the five conditions occupy non-overlapping regions of a 2-D manifold. We quantify this with 1-NN purity: pooling 1500 random top-layer hidden states from each of the five conditions and asking whether each sample's nearest neighbour (Euclidean, full 512-d state) comes from the same condition, the answer is yes for $7500/7500$ of them in our sample (1-NN purity $= 1.000$, vs. a chance baseline of 0.20). Within this 7500-sample test, the five hidden-state subspaces are not just probe-distinguishable: they are linearly separable to single-nearest-neighbour granularity. **[TODO: Whether this** 1.000 **holds at much larger sample size or under domain shift (MP3D-pooled, fov-shifted-pooled when available) is a further-investigate item.]**

Boundaries Our similarity tests are linear (CKA) or near-linear (t-SNE default); non-linear similarity measures could reveal higher-order shared structure. Our five sensors are diverse but not exhaustive; hybrid sensors (coarse uniform periphery + sharp fovea, depth-only inputs) might produce intermediate geometries.

Memory transplant Probe-based similarity measures could still miss *functional* equivalence: two representations with low CKA might still be interchangeable if the downstream policy reads them the same way. We test this with memory transplant — paste a donor's mid-episode hidden state into the recipient's policy at step t, let the recipient drive the rest of the episode. The isolated condition-mismatch cost (cross-transplant SPL $-$ self-transplant SPL, which subtracts both a small $t{=}0$ protocol-noise floor and the rollout-divergence cost of re-initialising a recurrent state mid-episode) is stable across midpoints $t \geq 15$ and carries a clear ranking (Figure 4b): foveated-learned$\rightarrow$foveated incurs the largest drop (-0.17 SPL), foveated$\rightarrow$uniform a smaller one (-0.10), and foveated$\rightarrow$blind is essentially neutral (-0.02). (The raw cross-transplant drop vs. baseline is correspondingly 0.19–0.21 for the first two pairs; self-transplant contributes a 0.02–0.04 SPL overhead that accounts for the difference.) The striking piece is the ranking itself: pairs that look identically near-zero in CKA differ markedly under transplant. Higher probe-based similarity is uninformative at this scale — representations that look the same to a linear similarity measure are not interchangeable in policy terms. (CKA and probe transfer use common-sample truncation for fair cross-condition comparison; transplant uses full on-policy rollouts because the task is not truncatable. The two measurement scopes are deliberate, and their conclusions are consistent.) Together with CKA, probe transfer, and t-SNE, transplant provides a fourth convergent signal of format-level divergence on the pairs we tested (foveated$\rightarrow$blind, foveated$\rightarrow$uniform, foveated-learned$\rightarrow$foveated). **[TODO: The**

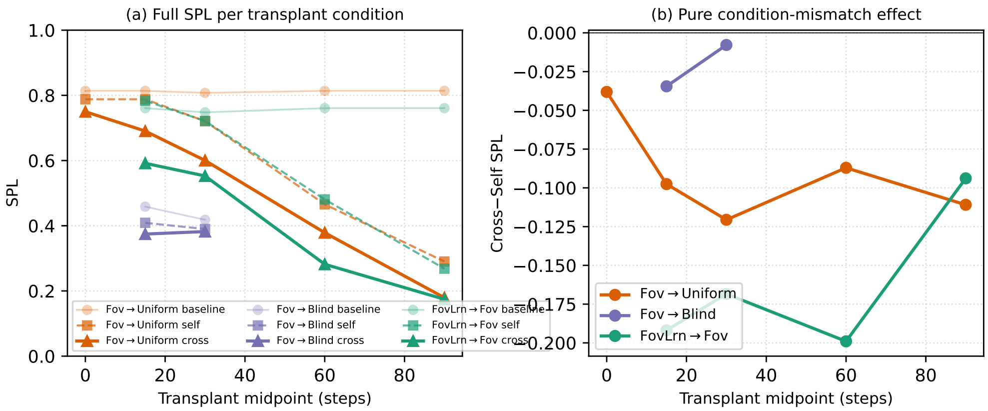


Figure 4: Memory transplant across a midpoint sweep $\{0, 15, 30, 60, 90\}$ steps ($n = 100$–150 episodes/cell). *(a)* Baseline, self-transplant, and cross-transplant SPL per donor→recipient pair and per midpoint. At $t = 0$ the donor has accumulated no information yet, so self- and cross-transplant reduce to a protocol-noise floor (self ≈ -0.03, cross ≈ -0.06 SPL vs. baseline). Self-transplant SPL drops monotonically with increasing midpoint: the recipient's own past recurrent state genuinely diverges from the state it would have occupied had it driven the rollout from step 0. *(b)* Cross minus self (the isolated condition-mismatch component, with both protocol noise and rollout divergence subtracted) is roughly stable across midpoints $t \geq 15$: foveated-learned→foveated is the most behaviourally incompatible pair at every midpoint tested (-0.17 SPL on average), even though all pairwise CKA values are at the noise floor. Probe-based similarity is uninformative at this scale; transplant is what reveals the functional gap.

remaining cross-condition transplant pairs (e.g., blind→matched, matched→uniform) are not yet measured; running them would let us state the format-divergence finding as a complete 5×5 behavioural matrix rather than a representative subset.]

4.4 H3: Gaze Location as an Orthogonal Content Lever

H1 and H2 identify sensor structure as one axis on representational content. H3 asks whether gaze *location* — where the foveated window is centred — is a second, dissociable axis. We test the two forms described in §3.3 and previewed in the Introduction: an active-gaze form (the minimal learned-gaze module should move its gaze to task-useful locations), and a gaze-location form (two foveated agents whose gaze is centred at *different static* locations should produce different content). The active-gaze form is out of scope of our minimal gaze decoder and we do not claim anything about it; the gaze-location form is what our controlled experiments test.

The learned-gaze module collapses Our minimal learned-gaze implementation (deterministic sigmoid MLP, slow-gaze updates once per 128-step rollout segment; Appendix B) does not produce active gaze: after 250M frames the gaze output is frozen at $(0.49, 0.62)$ with per-dimension std $\leq 2.5 \times 10^{-4}$ (numerical noise). Two anti-collapse regularisers (Appendix B) restore gaze variance only at the cost of task performance. We explicitly do not claim this shows active gaze cannot be learned — our gaze decoder is minimal by design (to be a controlled variable, not a model of active vision), and richer architectures (stochastic gaze, information-seeking intrinsic reward, attention-based token selection) would need separate investigation. What the collapse gives us for free is an opportunity to test the gaze-location form: after training, we have an agent that is *effectively foveated with a static gaze* 30 *pixels below image centre*.

Gaze-location test (in progress) Both rich-encoder foveated variants sit in the pass-through regime under current-state probing (Table 1), so the H3 test is whether *within* the pass-through regime, gaze location shifts content along other axes — non-spatial feature usage, behavioural memory-reliance (shortcut, Figure 5b), or transplant compatibility. To isolate gaze location from training-dynamics confounds, we train a foveated-shifted control: an agent identical to foveated-fixed except gaze is hardcoded at $(0.49, 0.62)$ instead of $(0.5, 0.5)$ — matching foveated-learned's collapsed gaze

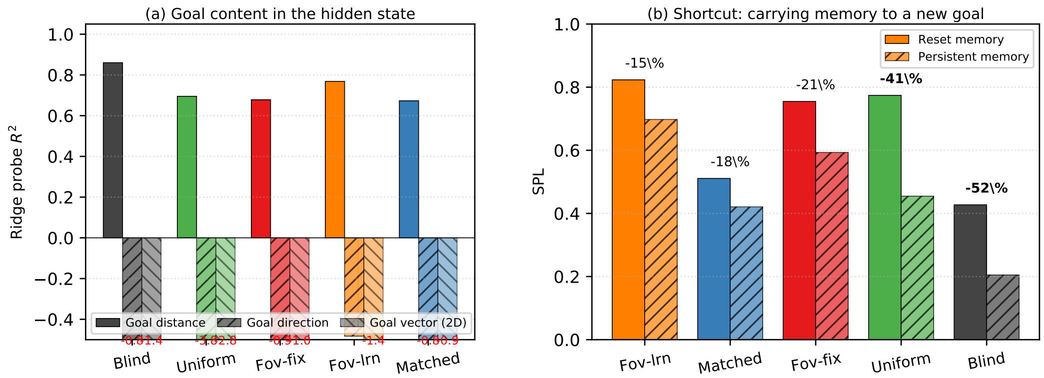


Figure 5: H3 content evidence. (a) Goal-vector probe R^2 (deterministic-rollout hidden states): goal *distance* decodes strongly across conditions, while goal *direction* and the full 2-D ego-to-goal vector are at chance or negative for every condition — no condition stores the goal vector explicitly (PointGoal provides it as a per-step input, so the policy has no incentive to re-encode it). What differs across conditions is the integrated trajectory and scene-history content each recurrent state accumulates. (b) Shortcut discovery: persistent vs. reset memory on the second of two paired same-scene/different-goal episodes. 20 Gibson scenes $\times$ 10 paired episodes each, bar height is the mean of the 20 scene-level mean SPLs (not a flat 200-sample mean, to prevent high-episode-count scenes from dominating). Conditions are ordered by relative SPL drop (%). Blind loses most; foveated-learned loses least.

position but with no learned-gaze module in the optimisation path. **[TODO: Training in progress at submission; results will populate this paragraph.]**

Shortcut discovery (behavioural memory-reliance) We separately measure how tightly each policy couples action choice to integrated recurrent memory: run paired episodes in the same scene with fresh goals, compare second-episode SPL under *reset* (LSTM zeroed between episodes) vs. *persistent* (state carried over) memory. Larger relative SPL drop = more brittle memory coupling (Figure 5b): blind loses 52%, uniform 41%, foveated 21%, matched 18%, foveated-learned 15%. The blind ranking is consistent with H1: blind has the strongest top-layer GPS code of the five conditions and is also the most brittle when that memory is stale; the three rich-encoder conditions, which decode GPS at chance from the top-layer state, are correspondingly less perturbed. *Note that "decodable from h" and "policy uses for decision-making" are not the same construct* — a probe shows what is *readable* from the state, not what the policy actually *relies on*; the shortcut experiment is the closer behavioural test of the latter. **[TODO: A direct measurement of "policy GPS reliance" (e.g., ablating GPS sensor input mid-rollout and measuring the SPL drop) would complement the shortcut signal; left for follow-up.]** *Anomaly: the matched-compute drop (18%) is lower than its strong GPS encoding might predict.* A literal reading is that *using* a cognitive map for policy decisions is a separate property from *having* one — matched has the code but weights it less in policy output — but this rests on a single-seed shortcut run; we mark it as a candidate dissociation pending multi-seed replication of both the GPS probe and the shortcut measurement before drawing the "having vs. using" conclusion.

4.5 Additional Analyses

Controls Two probes fail to discriminate between conditions and so define the boundary of our claims. Coarse occupancy ("have I visited this region?") decodes above chance in every condition, so one-bit spatial working memory is ubiquitous and not part of the H1–H3 story. Metric 20×20 occupancy reconstruction fails everywhere, consistent with the SPACE benchmark's finding that fine-grained layout requires non-linear probes or longer horizons [Ramakrishnan et al., 2025]: our claims are about representational *format* and *content*, not about high-resolution metric layout.

Per-unit spatial information and multi-layer depth Two further analyses sharpen the H1 picture (Figure 6). *Per-unit:* bottleneck conditions distribute spatial information broadly across LSTM units in the Banino et al. [Banino et al., 2018] sense (many units each carrying > 1 bit) rather than

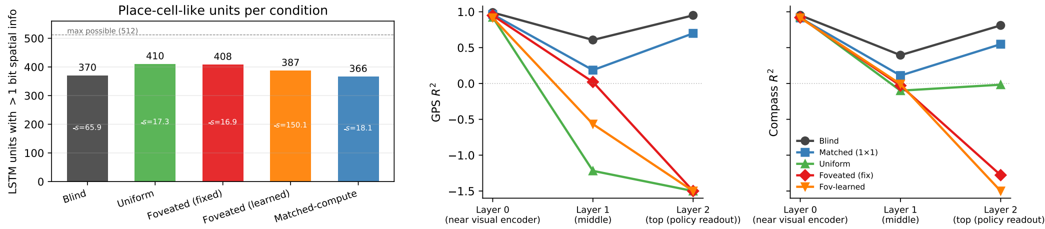


Figure 6: *Left*: LSTM units with > 1 bit spatial information per condition [Banino et al., 2018]; mean bits $\bar{s}$ shown inside bars. *Right*: Per-LSTM-layer GPS / compass probe R^2 across the three layers, all five conditions, on deterministic-rollout hidden states. All five conditions have linearly-decodable GPS at Layer 0 ($R^2 > 0.91$); Layer 0 is also where the GPS sensor embedding enters the LSTM input vector, so this lower-bound is at least partly a sensor-pass-through effect rather than "the network computing position." What differs across conditions is what the LSTM does with this signal through Layers 1–2: bottleneck conditions (blind, matched) retain it to the top-layer policy readout, while rich-encoder conditions (uniform, foveated, foveated-learned) discard it (top-layer R^2 catastrophically negative). Some non-monotonicity in blind's per-layer trace (Layer 1 lower than Layer 0 or Layer 2) is not yet explained.

concentrating it in a few dedicated "place neurons," consistent with a population code. *(Caveat: this depends on the chosen* 1*-bit threshold and the per-scene grid resolution; we do not claim a sharp dissociation between "population code" and "few dedicated cells.") Multi-layer:* probing every LSTM layer (not just the top) shows that all five conditions have linearly-decodable GPS at Layer 0 ($R^2 > 0.91$), where the GPS sensor embedding enters the LSTM input — so Layer 0 is at least partly a sensor-pass-through artefact, not by itself evidence that the network "computes" position. The interesting signal is the per-layer trajectory: bottleneck conditions *retain* the GPS code through Layers 1–2 to the top-layer policy readout, while rich-encoder conditions discard it (uniform Layer 2 GPS $R^2 = -1.56$, foveated -1.64, foveated-learned -10). An MLP probe on the same top-layer state recovers position from rich-encoder networks (foveated $R^2 = 0.51$, foveated-learned $R^2 = 0.73$), *consistent with* the position information persisting in a non-linear subspace — though we cannot from this alone exclude MLP exploitation of position-correlated features (wall distance, scene identifiers) rather than position itself. *(A non-linear-position-encoding test that holds the input-resolution constant while varying encoder capacity is the encoder-resolution scaling sweep, Appendix D, in progress.)* What is robust on current evidence: "linearly decodable at the top-layer policy readout" is the precise meaning of H1, and bottleneck and rich-encoder conditions diverge sharply on this metric. *(Side observation, not yet explained: blind Layer 1 GPS R^2 is lower (0.61) than at Layer 0 (0.99) or Layer 2 (0.95); the LSTM does not pass GPS through monotonically. We flag this as an artefact-or-mechanism question to be resolved in follow-up.)*

Generalisation to MP3D The H1 bottleneck ordering is preserved on a held-out MP3D evaluation (300 episodes per condition, deterministic-rollout protocol; same trained checkpoints, no re-training). Blind GPS R^2 moves from 0.95 on Gibson to 0.93 on MP3D ($\Delta = -0.02$); matched moves from 0.78 to 0.71 ($\Delta = -0.07$); the three rich-encoder conditions (uniform, foveated, foveated-learned) remain at chance or sub-chance on both datasets (Figure 7). The bottleneck-vs-pass-through partition is therefore not specific to Gibson scene statistics. *Anomaly:* fov-learned compass moves from -1.34 on Gibson to $+0.41$ on MP3D — a $+1.75$ swing on a single-seed checkpoint. We do not yet have a confirmed account; possibilities include Gibson-specific feature use by the fov-learned encoder, single-seed noise, or an interaction between fov-learned's collapsed gaze location and MP3D scene statistics. We mark this as an open observation requiring multi-seed replication and the fov-shifted causal control before claiming any specific mechanism. The bottleneck-vs-pass-through partition itself is unaffected.

5 Discussion

5.1 Two orthogonal axes: sensor structure and gaze location

Our findings unify under a single mechanism: the *encoder–memory race*. Whether the LSTM carries world-frame spatial structure depends on whether the visual encoder can resolve it from the current

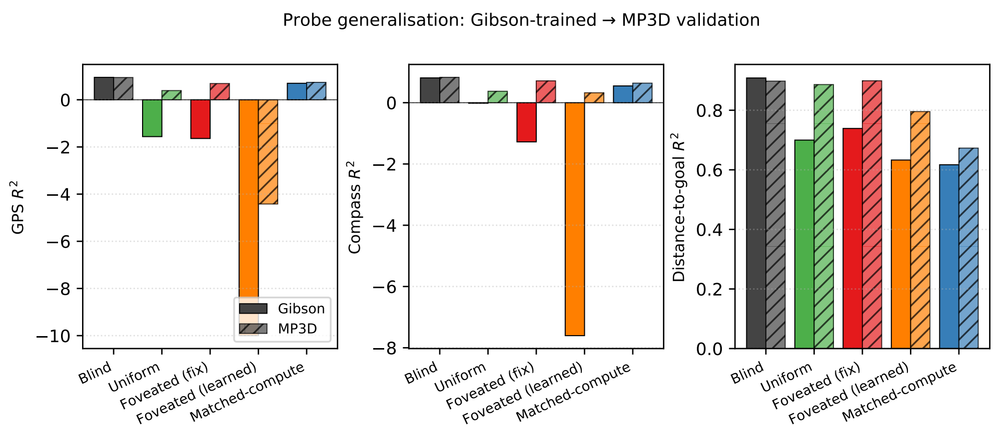


Figure 7: Probe generalisation from Gibson evaluation (unshaded) to held-out MP3D evaluation (hatched), 300 episodes per condition under the deterministic-rollout protocol. Same trained checkpoints; no re-training. The H1 bottleneck ordering — blind and matched-compute decode world-frame GPS robustly while uniform, foveated, and foveated-learned do not — is preserved across the dataset shift. Foveated-learned compass shows the largest cross-dataset shift (Gibson $-1.34 \rightarrow$ MP3D $+0.41$); we report this as an open anomaly requiring multi-seed replication before attributing it to a specific mechanism. The qualitative bottleneck partition is unaffected.

frame. When the encoder cannot — blind (no encoder) or matched-compute (1×1 feature map) — the LSTM takes over and maintains a persistent cognitive-map-like code across many time-steps of lag (H1, §4.2). When the encoder can — uniform, foveated, foveated-learned all use full-resolution ResNet-18 encoders — the top-layer LSTM state drifts to pass-through, even though all conditions retain linearly-decodable GPS at Layer 0 (where the GPS sensor embedding enters the LSTM input; the divergence is at Layers 1–2, Figure 6 right). H2 (§4.3) cuts orthogonally: *what* the LSTM encodes is in different linear subspaces across conditions even where current-state position decodes at chance, and that linear difference matters for downstream behaviour (cross-condition memory transplants drop SPL more than transplant mechanics alone). H3 (§4.4) asks whether, *within* the rich-encoder pass-through regime, gaze location is a second representational-content axis; the clean foveated-shifted causal control that answers this is in training at submission time. The same encoder-capacity axis that collapses the original "foveated vs. uniform" question (both sit in the pass-through regime) surfaces a clean scaling prediction: GPS decodability should fall monotonically with encoder resolution, which we begin to test in Appendix D.

5.2 Why this, not alternative accounts

We rule out four alternative explanations.

Task-competence differences. Every condition solves PointGoal at 93–99% success, so the encoded-content gap cannot be attributed to differences in navigation skill (Table 1).

Probe-capacity / probe-fitting artefacts. DtG decodes robustly across all five conditions ($R^2 \geq 0.81$), demonstrating that Ridge has the capacity to fit high-R^2 scalars when the hidden state contains them. Where GPS does decode (blind, matched), Hewitt–Liang label-permutation selectivity is correspondingly high; where GPS is at chance, selectivity is also at chance because real and control probes face the same near-zero target-information ceiling. The chance-level GPS/compass R^2 for rich-encoder conditions is therefore a property of the hidden state, not of the probe.

Sample-size / trajectory-distribution artefacts. CV σ spans come from the same 500-episode pool with $\sim$100–400-step rollouts under identical deterministic evaluation; the protocol confound that haunted an earlier draft (4-step quasi-static trajectories under stochastic sampling) is documented and retired in §3.3.

Linear-probe artefacts. Two complementary checks address this. *(a) Behavioural:* the memory-transplant intervention (§4.3) goes outside the probe framework and reproduces the bottleneck-vs-pass-through ordering, with cross-condition transplants dropping SPL by 0.19–0.21 between rich-encoder pairs. *(b) Non-linear probe:* a 2-layer MLP on the same top-layer hidden states recovers $R^2 \in [0.51, 0.73]$ from the rich-encoder conditions where Ridge sees chance. We interpret this as consistent with position information persisting in a non-linear subspace at the top layer, but emphasise the limits: an MLP probe can also exploit features merely *correlated* with position (wall distance, scene identifiers, episode-phase markers), so this number alone cannot disentangle "non-linear position code" from "MLP regressing through position-correlated features." What it does establish is that the bottleneck-vs-pass-through divergence is not just about whether *any* probe can recover position — it is specifically about whether a *linear* top-layer probe can. Since the standard DD-PPO action head is a linear projection from $\mathbf{h}_2$, the linear-probe metric is precisely what the policy itself can use — so the H1 distinction maps directly onto policy-accessible encoding regardless of how the MLP-probe interpretation resolves. The encoder-resolution scaling sweep (Appendix D, in progress) tests the bottleneck framing more directly by varying input resolution while holding everything else fixed.

5.3 Biological precedent as convergent evidence

Each of our three findings has a parallel in the animal-navigation literature, which we take as one piece of convergent (not conclusive) evidence that the effects are not specific to a single LSTM architecture. Compensatory memory under reduced sensory access is documented in congenitally blind humans, who develop enhanced hippocampal coupling and outperform sighted controls on non-visual spatial-working-memory tasks [Kupers and Ptito, 2014], and in rodents whose grid-cell periodicity collapses in darkness despite intact head-direction inputs [Chen et al., 2016]—both format-level, not just strength-level, effects. Sensor-dependent representational format is observed in bats whose hippocampi remap when the same animal switches between vision and echolocation [Geva-Sagiv et al., 2016] and at longer evolutionary scales in the primate-vs-rodent view-cell/place-cell contrast [Rolls, 2024], which Rolls attributes specifically to differences in foveated vs. near-panoramic sampling statistics. Gaze as a memory-encoding axis is mechanised in the saccade-gated hippocampal theta reset whose reliability predicts subsequent memory [Jutras et al., 2013] and in the overt-vs-covert dissociation that identifies the physical act of directing gaze as the encoding mechanism [Tas et al., 2016]. We do not claim our agents implement these biological mechanisms, nor that the parallel structure settles the generality question; we do claim that the parallels make a pure single-architecture artefact less likely, and that the controlled agent setting complements biological comparisons where sensor type, architecture, and training objective cannot be independently manipulated.

5.4 Implications for ML practice and architectural scope

Within the recurrent PointNav setting we study, three implications follow. (i) *Encoder capacity correlates inversely with linear, top-layer, policy-accessible LSTM spatial encoding* across our five conditions. If the correlation reflects causation in this setting — which the encoder-resolution scaling sweep (Appendix D, in progress) tests directly — it would also be a practical training-time lever for inducing cognitive-map-like internal representations: impose an encoder bottleneck. We do not yet claim this generalises beyond PointNav with our specific architecture and training regime. (ii) *"More pixels" is not the right lens within this setting:* the linear top-layer GPS code is non-monotonic in raw pixel count — uniform 256×256 (full encoder) lacks it; 48×48 with the same encoder stack (matched-compute, 1×1 feature map) has a strong one. We attribute this to encoder-preserved spatial information, with the scaling sweep needed to confirm. (iii) *Attention policy (H3, pending):* within the rich-encoder regime, we predict that shifting the static gaze location alters what the LSTM encodes; the foveated-shifted causal control is in training at submission time and will test this directly (§4.4).

Beyond the recurrent setting (untested) We use a Wijmans-style recurrent (LSTM) backbone for direct comparability with the emergent-map literature. The three implications above are established only in this setting, but we speculate about where they plausibly extend. Transformer-based embodied agents build their internal state by accumulating information via learned attention, which is structurally similar to our gaze location in the sense that it selects which observation content is read into the representation; our H3 finding therefore suggests a testable prediction for transformer navigators—strong constraints on attended tokens should produce content-level dissociations of the sort we observe—but the prediction has not been verified and is left for future work. The sensor-structure axis (H1, H2) is a claim specifically about RL recurrent navigation agents trained under partial observability; whether equivalent representational-content effects arise in supervised visual learners, non-visual modalities, or multimodal systems is an open empirical question that the present experiments do not resolve.

5.5 Limitations

(i) *Intermediate checkpoint for foveated-fixed:* all foveated-fixed analyses use ckpt.36 at $\sim$174M, the last checkpoint before the silent-corruption window (Appendix C); the other four conditions use the final converged checkpoint. (ii) *Minimal learned-gaze architecture.* Our learned-gaze module is a deterministic sigmoid MLP with slow-gaze rollout updates, chosen for computational compatibility with DD-PPO rollouts and simplicity as a controlled variable, not as a competitive active-gaze model. Its collapse under pure task supervision is specific to this architecture (and so far to a single seed; multi-seed replication is in progress); richer gaze mechanisms (stochastic gaze with entropy bonus, information-seeking intrinsic reward, attention-based token selection) would need separate investigation. We therefore do not claim that implicit task pressure cannot produce active gaze in general; we claim only that our minimal decoder collapsed, and we use the collapsed-gaze position as an empirical starting point for the gaze-location test (foveated-shifted control, §4.4) — a test whose logic does not depend on the gaze decoder working. (iii) *Cross-heading probe:* the proposal included a cross-heading generalisation probe (train on heading A, test on opposite heading). Our implementation returned "insufficient samples" sentinels at our probe sizes and does not appear in our results. (iv) *Foveation model:* Gaussian blur with quadratic falloff is a simplification; log-polar or multi-scale pyramid models may behave differently. (v) *Linear probes only:* non-linear probes might reveal structure Ridge regression misses, particularly for occupancy maps. (vi) *Architecture scope:* we use a recurrent (LSTM) backbone to remain comparable with the emergent-map literature we build on. The core claims (sensor structure and gaze location as representational-content axes) should hold for transformer-based navigation agents whose internal state is built by learned attention, but directly verifying this requires re-running the five-condition ablation on a transformer backbone and is left for future work; see §5 for the specific predictions.

6 Conclusion

We ran a controlled five-condition ablation isolating the effect of visual-input structure on the recurrent representations learned by an otherwise-identical DD-PPO navigation agent. Three findings emerge.

(1) Encoder–memory race. LSTM world-frame spatial encoding monotonically tracks visual-encoder information capacity. Blind and matched-compute (the conditions whose encoder cannot resolve world-frame structure) develop persistent cognitive-map-like codes that decode at $R^2 = 0.95$ and 0.78 respectively at $k = 0$, hold across ≥ 10 time-steps of lag, and reproduce on a held-out MP3D evaluation. The three rich-encoder conditions (uniform, foveated, foveated-learned) decode at chance from the top-layer state, although per-layer probing shows all five conditions have linearly-decodable GPS at Layer 0 ($R^2 > 0.91$) where the sensor embedding enters — the divergence is what happens through Layers 1–2. An MLP probe still recovers position at Layer 2 in rich-encoder conditions ($R^2 \in [0.51, 0.73]$), consistent with the information persisting in a non-linear subspace; we cannot from this alone exclude MLP exploitation of position-correlated features. H1 is most precisely a claim about what a *linear, top-layer* probe can read — which is also what the standard DD-PPO action head can read.

(2) Condition-specific representational format. Even where rich-encoder LSTMs are spatially pass-through, their hidden-state subspaces are perfectly disjoint: pairwise CKA at the noise floor ($< 10^{-4}$), cross-condition probe transfer catastrophically negative ($R^2 \ll -800$), 1-NN purity 1.000 across pooled samples, and cross-condition memory transplants that drop SPL by 0.19–0.21 between rich-encoder pairs (isolation metric up to -0.17). Whatever non-spatial task-relevant features each LSTM carries, it carries them in its own linear subspace — so much so that a probe trained on one condition predicts off-manifold values on another.

(3) Gaze-location test (pending). Whether gaze location is a second representational-content axis *within* the rich-encoder regime is what H3 asks; the cleanest test (foveated-shifted training with hardcoded gaze at the learned-gaze module's collapsed location) is in progress at submission and will report in §4.4.

The pattern parallels three threads of animal-navigation research — enhanced hippocampal coupling in congenitally blind humans, the disruption of rodent grid-cell firing in darkness with navigation nonetheless preserved, and cross-modality hippocampal remapping in bats. We use these as analogy only: the convergent direction across systems is suggestive of a sensor–memory coupling principle, but our experiments do not test architectures other than LSTM, so we do not claim the principle is architecture-independent. Open questions: (i) whether the encoder-bottleneck ordering extends to finer-grained resolution sweeps (Appendix D); (ii) whether H3 stands on the clean causal test; (iii) whether transformer-based navigation agents reproduce the encoder–memory race; (iv) whether equivalent effects shape representations in non-navigation embodied tasks; (v) multi-seed replication of all five conditions to bound seed-to-seed variability of the H1 numbers.

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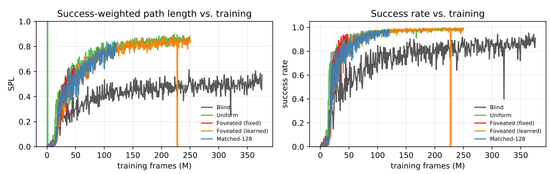
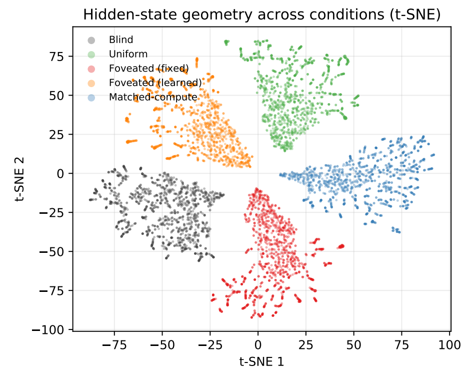


(a) Training curves (TensorBoard event-file scalars). All four sighted conditions converge to SPL 0.82–0.85 and success $> 97\%$ by $\sim$100M frames; blind converges to SPL $\sim$0.50 over a longer horizon. The glitch at frame $\approx$ 228M (fov-learned) is a resume-boundary artefact in the scalar log.



(b) t-SNE projection of top-layer LSTM hidden states (deterministic-rollout probes), pooled across all five conditions (perplexity 30). The five conditions occupy non-overlapping regions of the 2-D embedding — a visual confirmation of the near-zero CKA result (§4.3).

Figure 8: Supplementary visualisations referenced from the main text.

B Gaze-Diversity Auxiliary Loss (H3 Ablation)

Motivation In the first foveated-learned training run the gaze decoder collapsed (per-dimension std $\approx 5 \times 10^{-4}$). We attribute this to two mechanisms: (i) once the sigmoid output saturates, its derivative $\sigma'(x) \to 0$ attenuates gradient signal; (ii) consistent-gaze views yield more predictable visual features, so the reward-gradient path has no explicit incentive for variety.

Target-variance hinge We add $\mathcal{L}_{\text{div}} = \lambda \cdot \text{relu}(\sigma^2_{\text{target}} - \text{Var}_{\text{batch}}[g])$ where $g \in [0,1]^2$, $\text{Var}_{\text{batch}}$ averaged over the two gaze dimensions, $\sigma^2_{\text{target}} = 0.01$ (std ≈ 0.1), $\lambda = 0.01$. The hinge disappears once variance exceeds the target. We also evaluated an un-capped $-\lambda \cdot \text{Var}[g]$ variant as a negative control.

Pilot 1 (uncapped, job 2840256) After 10 M frames, gaze variance exceeded that of a uniform distribution over $[0,1]^2$ (std $[0.43, 0.42]$ vs. theoretical max 0.289): gaze was essentially random. Task performance collapsed: SPL plateaued at 0.05, running-mean metrics transitioned to NaN at ~ 8 M frames. The uncapped regulariser destroyed the consistent-foveation signal the policy needs.

Pilot 2 (hinge, job 2842288) Gaze converged to an extreme corner of image space: mean $(0.001, 0.998)$ with std $[0.020, 0.023]$ — well below the target std of 0.1 and therefore with the hinge active throughout. Task SPL plateaued at 0.05 as in pilot 1. The policy apparently found the minimum-task-cost gaze that partially satisfies the diversity pressure: a sigmoid saturated at one corner, producing a constant image-corner view.

Interpretation Both pilots argue that in our deterministic sigmoid-MLP gaze architecture with slow-gaze rollouts, gaze collapse is a pre-requisite for task learning rather than an accident. Task signal alone does not spontaneously produce informative fixation, and an exogenous anti-collapse regulariser moves gaze toward whatever region reduces regulariser loss fastest, which is generically *not* the task-useful region. A genuine H3 test likely requires (i) a stochastic gaze location with entropy bonus, so diversity is sampled at rollout time rather than enforced on a deterministic output; or (ii) a task-aligned intrinsic reward (e.g., information gain about goal-relative position); or (iii) an architectural change separating gaze generation from visual-feature extraction (e.g., attention-based gaze).

C Training Stability: Silent Weight Corruption in DD-PPO

During an initial foveated training run we observed a failure mode that did not manifest as a crash but as slow, silent corruption of the learned weights. After ≈170M frames of stable progress (reward ≈ 10, SPL ≈ 0.83), one mini-batch on one DDP worker produced a NaN gradient. Because `torch.nn.utils.clip_grad_norm_` does not sanitise non-finite gradients—the global norm becomes NaN, the clip coefficient becomes NaN, and `grad *= NaN` remains NaN—the subsequent `optimizer.step()` wrote NaN into every parameter. DDP all-reduce propagated the corruption across workers. The training loop did not raise: it continued for 2.5 days with reward=NaN and SPL$= 0$ until `torch.multinomial` finally rejected the NaN probability tensor. All checkpoints saved during this window were unrecoverable (90 of 97 parameter tensors entirely NaN).

To isolate the origin of the NaN gradient we re-ran training from the last clean checkpoint with `torch.autograd.set_detect_anomaly(True)` and per-module forward-output hooks. The anomaly traceback pinned the failure to `MseLossBackward0`, i.e. the backward pass of the value-head loss $\frac{1}{2}(V(s) - R)^2$; rare return-vs-value discrepancies overflowed float32 in the squaring step, producing non-finite gradients downstream.

Our mitigation is a minimally invasive patch to `PPO.before_step` that zeroes any non-finite gradient element before clipping. A bad mini-batch becomes a no-op update rather than a catastrophic one. The patch is a no-op on clean training paths. Across the five conditions reported here, no NaN events were observed during full training after deployment, and all final checkpoints contain only finite weights. We reported the underlying issue upstream (habitat-lab issue #2226) with a minimal reproduction and a proposed native fix.

D Encoder-capacity scaling sweep

[TODO: The scaling sweep trains matched-compute variants at input resolutions $\{32, 48, 64, 96, 128, 192\}$ alongside the existing blind (no encoder) and uniform 256×256 anchors. Probe results on the trained checkpoints will populate Figure 9 and this appendix section. Hypothesis: GPS probe R^2 decreases monotonically with input resolution, from $R^2 \approx 0.95$ at no-encoder to $R^2 \approx 0$ at 256×256, crossing into the pass-through regime between 48 and 128 pixels.]

Figure 9: Encoder-capacity scaling curve. **[TODO: Pending data.]** x-axis: input resolution in pixels; y-axis: GPS and compass probe R^2 (5-fold CV, mean$\pm\sigma$).